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GNIOT Institute of Management Studies – PGDM (Batch 2021-23)

(TRIMESTER – II) SUPPLEMENTARY EXAMINATION, MAY - 2023

Operation Management

[Time: 2 Hours 30 Minutes]

[Maximum Marks: 100]

INSTRUCTIONS:

- (i) **There are THREE sections in this question paper.**
- (ii) **Attempt questions from each section as per the directions.**
- (iii) **Make suitable assumptions wherever necessary.**

SECTION-A

(Short Answer Type Questions, do not exceed 100 words)

Q.1. Attempt all parts of the following:

(10×3=30)

- (a) What is the various type of inventory cost used in inventory management?
- (b) Explain the difference breakdown maintenance and preventive maintenance with the help of examples.
- (c) “Supply chain management is the integrated management of all the linkages and value-added activities form the supplier-organization customer”. Discuss
- (d) What do you mean by the term ‘Production planning and control’? Explain briefly the objectives of PPC.
- (e) Discuss why operation management is more important than other department with the help of example of an industry.
- (f) What is the basic difference between production and operation management?
- (g) Explain various techniques of Work measurement with the help of examples
- (h) Explain the concepts of Economic Order Quantity with the help of examples
- (i) What do you mean by project management and discuss various functions of a project manager?
- (j) Explain the concepts of TQM with the help of examples.

SECTION-B

(Long Answer Type Questions)

Attempt all questions of the following:

(4×10=40)

Q.2. (a). What is the major scope of operation management that manufacturing organizations face in the country?

OR

(b) Battery Wholesale, Inc., purchase batteries for Rs. 14 each and it cost Rs 11 to process an order. The company sells about 12,000 of a particular type of battery per year at a uniform rate. The order lead time is 3 days. The holding cost is estimated to be 24 percent of the item cost per year. Determine the EOQ, The total cost and the reorder level.

Q.3. (a) One of the strategies for being cost effective is the location of the Plant. What factors would you consider while locating an automobile plant? Explain the priority you would assign for each factor.

OR

(b) Define the term plant layout. Discuss various types of layouts with the help of examples.

Q.4. (a) 8 hours work measurement study in a plant reveals the following. Using produced = 320 No, Idle time = 15%. Performance rating = 120%. Allowance = 12% of Normal time. Determine the standard time per unit produced. Also explain the difference between method study and work measurement.

OR

(b) What measures of quality of output do you think might be important in the following kinds of system?

- (i) A fast-food operation such as McDonalds
- (ii) A luxury hotel

Q.5. (a) A manufacturer of farm equipment is considering three locations A, B, and C for a new plant. Cost per year at the site are Rs 2,40,000, Rs 2,70,000 and Rs 2,52,000 respectively whereas variable costs are Rs 100 per unit, Rs 90 per unit and Rs 95 per unit respectively. If the plant is designed to have an effective system capacity of 2500 unit per year and is expected to operate at 80% efficiency. What is the most favourable location on the basis of actual output?

OR

(b) Explain the difference between work study, method study and work measurement with the help of examples.

SECTION-C

Q.6. Attempt all parts of the following:

(2×15=30)

6(a) Study the situation of inventory system of Johnson and Company and answer the question.

JOHNSON & CO.

The Company's Sales Department has opened up four new warehouses at different locations to speed up customer services and the last one was made operational just before the peak season started.

The results were, however, not as anticipated and the sales dropped by 10% to 15% in each district, partially due to lack of experience in running warehouses and also due to improper coordination between plant and warehouse.

Mr. Jacob Johnson, the president of the company, on the other hand had been informed by Mr. Smith, the Inventory Manager, that due to incorporation of new warehouses, the company is forced to carry huge inventories. Mr. Johnson was very upset due to increased cost and less sales, and therefore called for details from Mr. Parry, the Sales Manager.

Mr. Parry, upset himself with loss in sales and failure of warehousing system, was unable to understand the complaint from Mr. Johnson and apprised him that the plant did not cooperate with warehouses and was still supplying nearly 40% of total orders to customers directly from plant. This has resulted in demoralization of sales force and wanted Mr. Johnson's interference in the matter so as to support the field staff by increasing the supplies to warehouse. He also clarified that the total inventories of four warehouses and one at central location was still at the same level as before and the total inventory has not increased.

The report from Mr. Parry was then forwarded to Mr. Smith for his detailed comments and proposed action plan, so that field staff is satisfied and inventory level is maintained satisfactorily. Mr. Smith reverted to Mr. Johnson stating that warehousing operation was not scientifically managed, since they were not sending any advance information regarding the requirements. The orders of customers for shortage items always got precedence over warehouse orders but the warehouse orders were in any case dispatched within 2 to 3 weeks. He also confirmed that although all the parts were available between the central location and four warehouses combined, the shortage occurred at different warehouse. The plant was forcing him to carry huge inventory because it had to work on a steady rate even during slack period, while the warehouses were keeping very sleek inventory levels during slack period to boost high inventory turnover.

Therefore, Mr. Parry suggested dispatch of 3 months' produce to central location and four warehouses to avert the dissatisfaction of sales department and also install a costly closed loop Computer Network between four warehouses and central location so that contingencies could be attended to.

- (a)** Mr. Johnson requires your recommendation as an expert consultant on Inventory System to take care of his inventory problem.
- (b)** What are the main issues in the case study and how to resolve that issue.